

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Database Management Systems
COURSE CODE: CSA0506

COURSE OUTCOME COVERED

CO2: *Apply the relational model, relational algebra, SQL query structures, and views to solve database querying and data manipulation problems. (BL3, BL6)*

Activity Tool : Analytical Problem Solving (High)
Assessment Tool Weightage: 35%

Dr. Hemavathi R

QUESTIONS			Total Marks: 50
Q.No	Question	Bloom's Level	Marks
1	Apply relational algebra operations (σ , π , $\cup$, $-$, $\times$, $\bowtie$) on the given Employee and Department relations to retrieve required records.	BL3 – Apply	5
2	Write SQL DDL statements to create STUDENT(SID, Name, DOB, DeptID) and DEPARTMENT(DeptID, DeptName) tables with all appropriate constraints.	BL3 – Apply	5
3	Formulate SQL queries using GROUP BY, HAVING, and aggregate functions (COUNT, SUM, AVG, MAX, MIN) on a given Sales dataset.	BL3 – Apply	5
4	Write a correlated sub-query to find employees earning more than the average salary of their own department.	BL3 – Apply	5
5	Apply all types of JOIN operations (INNER, LEFT OUTER, RIGHT OUTER, FULL OUTER, CROSS) on Employee and Project tables.	BL3 – Apply	5
6	Create a view 'DeptSalaryView' that displays department name, total employees, and average salary. Write a query on this view.	BL6 – Create	5
7	Construct a relational algebra expression for the following: Find names of students who are enrolled in all courses offered.	BL6 – Create	5
8	Write SQL DML statements (INSERT, UPDATE, DELETE) with	BL3 – Apply	5

	integrity constraint enforcement for a given scenario.		
9	Apply tuple relational calculus to express a query: Find all employees who work in the 'IT' department and earn more than 50000.	BL3 – Apply	5
10	Design a relational schema for an Airline Reservation System and write 5 SQL queries demonstrating joins, sub-queries, and views.	BL6 – Create	5

Code of Conduct:

I Sree(192524023) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

- I. Apply relational algebra operations (σ , π , $\cup$, $-$, $\times$, $\bowtie$) on the given Employee and Department relations to retrieve required records.

Assume:

Employee(EmpID, EmpName, DeptID, Salary)

EmpID	EmpName	DeptID	Salary
101	Rahul	D1	60000
102	Priya	D2	45000
103	Amit	D1	55000

Assume:

Employee(EmpID, EmpName, DeptID, Salary)

DeptID	DeptName
D1	IT
D2	HR

(a) Selection (σ)

Employees earning more than 50000

$\sigma_{Salary > 50000}(Employee)$

(b) Projection (π)

Display employee names only

$\pi_{EmpName}(Employee)$

(c) Union ($\cup$)

Employees from IT and HR

$ITEmployees \cup HREmployees$

(d) Set Difference ($-$)

Employees not working in IT

$Employee - \sigma_{DeptID = 'D1'}(Employee)$

(e) Cartesian Product ($\times$)

$Employee \times Department$

(f) Join ($\bowtie$)

Employee details with department names

$Employee \bowtie_{Employee.DeptID = Department.DeptID} Department$

2. Write SQL DDL statements to create STUDENT(SID, Name, DOB, DeptID) and DEPARTMENT(DeptID, DeptName) tables with all appropriate constraints

```
mysql> SELECT * FROM Department;
```

DeptID	DeptName
5	Civil
1	Computer Science
3	Electronics
2	Information Technology
4	Mechanical

```
5 rows in set (0.008 sec)
```

```
mysql> ^C
```

```
mysql> SELECT * FROM Student;
```

SID	Name	DOB	DeptID
101	Rahul	2004-03-15	1
102	Priya	2003-11-20	2
103	Amit	2004-01-10	1
104	Sneha	2003-08-05	3
105	Kiran	2004-06-25	5

```
5 rows in set (0.019 sec)
```

3. Formulate SQL queries using GROUP BY, HAVING, and aggregate functions (COUNT, SUM, AVG, MAX, MIN) on a given Sales dataset.

```
mysql> SELECT Product,COUNT(*) AS TotalSales
-> FROM Sales
-> GROUP BY Product;
+-----+-----+
| Product | TotalSales |
+-----+-----+
| Laptop  |          3 |
| Mouse   |          2 |
| Keyboard|          2 |
| Monitor |          1 |
+-----+-----+
4 rows in set (0.071 sec)

mysql> SELECT Product,SUM(Amount) AS TotalAmount
-> FROM Sales
-> GROUP BY Product;
+-----+-----+
| Product | TotalAmount |
+-----+-----+
| Laptop  | 150000.00 |
| Mouse   |  2500.00 |
| Keyboard|  4500.00 |
| Monitor | 12000.00 |
+-----+-----+
4 rows in set (0.012 sec)

mysql> SELECT Product,AVG(Amount) AS AverageAmount
-> FROM Sales
-> GROUP BY Product;
+-----+-----+
| Product | AverageAmount |
+-----+-----+
| Laptop  | 50000.000000 |
| Mouse   | 1250.000000 |
| Keyboard| 2250.000000 |
| Monitor | 12000.000000 |
+-----+-----+
4 rows in set (0.084 sec)

mysql> SELECT MAX(Amount) AS HighestSale
-> FROM Sales;
+-----+
| HighestSale |
+-----+
| 55000.00 |
+-----+
1 row in set (0.271 sec)

mysql> SELECT MIN(Amount) AS LowestSale
-> FROM Sales;
+-----+
| LowestSale |
+-----+
| 1000.00 |
+-----+
1 row in set (0.012 sec)

mysql> SELECT Product,SUM(Amount) AS TotalAmount
-> FROM Sales
-> GROUP BY Product
-> HAVING SUM(Amount)>50000;
+-----+-----+
| Product | TotalAmount |
+-----+-----+
| Laptop  | 150000.00 |
+-----+-----+
```

4. Write a correlated sub-query to find employees earning more than the average salary of their own department.

```
mysql> SELECT *  
-> FROM Employee E  
-> WHERE Salary >  
-> (  
-> SELECT AVG(Salary)  
-> FROM Employee  
-> WHERE DeptID=E.DeptID  
-> );
```

EmpID	EmpName	DeptID	Salary
101	Rahul	1	60000
104	Sneha	2	50000
105	Kiran	3	70000

3 rows in set (0.379 sec)

5. Apply all types of JOIN operations (INNER, LEFT OUTER, RIGHT OUTER, FULL OUTER, CROSS) on Employee and Project tables.

```

--> ON Employee.ProjectID=Project.ProjectID;

```

EmpID	EmpName	DeptID	Salary	ProjectID	ProjectID	ProjectName
101	Rahul	1	60000	1	1	AI Project
102	Priya	2	45000	2	2	Web Portal
103	Amit	1	55000	1	1	AI Project
104	Sneha	2	50000	3	3	Cloud System
106	Ravi	3	65000	2	2	Web Portal

5 rows in set (0.010 sec)

```

mysql> SELECT *
--> FROM Employee
--> LEFT JOIN Project
--> ON Employee.ProjectID=Project.ProjectID;

```

EmpID	EmpName	DeptID	Salary	ProjectID	ProjectID	ProjectName
101	Rahul	1	60000	1	1	AI Project
102	Priya	2	45000	2	2	Web Portal
103	Amit	1	55000	1	1	AI Project
104	Sneha	2	50000	3	3	Cloud System
105	Kiran	3	70000	NULL	NULL	NULL
106	Ravi	3	65000	2	2	Web Portal

6 rows in set (0.013 sec)

```

mysql> SELECT *
--> FROM Employee
--> RIGHT JOIN Project
--> ON Employee.ProjectID=Project.ProjectID;

```

EmpID	EmpName	DeptID	Salary	ProjectID	ProjectID	ProjectName
103	Amit	1	55000	1	1	AI Project
101	Rahul	1	60000	1	1	AI Project
106	Ravi	3	65000	2	2	Web Portal
102	Priya	2	45000	2	2	Web Portal
104	Sneha	2	50000	3	3	Cloud System
NULL	NULL	NULL	NULL	NULL	4	Mobile App

6 rows in set (0.373 sec)

```

mysql> SELECT *
--> FROM Employee
--> LEFT JOIN Project
--> ON Employee.ProjectID=Project.ProjectID
--> UNION
--> SELECT *
--> FROM Employee
--> RIGHT JOIN Project
--> ON Employee.ProjectID=Project.ProjectID;

```

EmpID	EmpName	DeptID	Salary	ProjectID	ProjectID	ProjectName
101	Rahul	1	60000	1	1	AI Project
102	Priya	2	45000	2	2	Web Portal
103	Amit	1	55000	1	1	AI Project
104	Sneha	2	50000	3	3	Cloud System
105	Kiran	3	70000	NULL	NULL	NULL
106	Ravi	3	65000	2	2	Web Portal
NULL	NULL	NULL	NULL	NULL	4	Mobile App

7 rows in set (0.222 sec)

```

mysql> SELECT *
--> FROM Employee
--> CROSS JOIN Project;

```

EmpID	EmpName	DeptID	Salary	ProjectID	ProjectID	ProjectName
101	Rahul	1	60000	1	4	Mobile App
101	Rahul	1	60000	1	3	Cloud System
101	Rahul	1	60000	1	2	Web Portal
101	Rahul	1	60000	1	1	AI Project
102	Priya	2	45000	2	4	Mobile App
102	Priya	2	45000	2	3	Cloud System
102	Priya	2	45000	2	2	Web Portal
102	Priya	2	45000	2	1	AI Project
103	Amit	1	55000	1	4	Mobile App
103	Amit	1	55000	1	3	Cloud System
103	Amit	1	55000	1	2	Web Portal
103	Amit	1	55000	1	1	AI Project
104	Sneha	2	50000	3	4	Mobile App
104	Sneha	2	50000	3	3	Cloud System
104	Sneha	2	50000	3	2	Web Portal
104	Sneha	2	50000	3	1	AI Project
105	Kiran	3	70000	NULL	4	Mobile App
105	Kiran	3	70000	NULL	3	Cloud System
105	Kiran	3	70000	NULL	2	Web Portal
105	Kiran	3	70000	NULL	1	AI Project
106	Ravi	3	65000	2	4	Mobile App
106	Ravi	3	65000	2	3	Cloud System
106	Ravi	3	65000	2	2	Web Portal
106	Ravi	3	65000	2	1	AI Project

24 rows in set (0.015 sec)

6. Create a view 'DeptSalaryView' that displays department name, total employees, and average salary. Write a query on this view.

```
mysql> SELECT *
      -> FROM DeptSalaryView;
```

DeptName	TotalEmployees	AverageSalary
Computer Science	2	57500.0000
Information Technology	2	47500.0000
Electronics	2	67500.0000

3 rows in set (0.216 sec)

7. Construct a relational algebra expression for the following: Find names of students who are enrolled in all courses offered.

```
mysql> Select * From Enrollment;
```

SID	CID
101	1
102	1
103	1
104	1
101	2
103	2
105	2
101	3
102	3
103	3

10 rows in set (0.011 sec)

8. Write SQL DML statements (INSERT, UPDATE, DELETE) with integrity constraint enforcement for a given scenario.

```
mysql> Select * From Student;
```

SID	Name	DOB	DeptID
101	Rahul	2004-03-15	1
102	Priya	2003-11-20	2
103	Amit	2004-01-10	1
104	Sneha	2003-08-05	3
105	Kiran	2004-06-25	5

5 rows in set (0.008 sec)

9. Apply tuple relational calculus to express a query: Find all employees who work in the 'IT' department and earn more than 50000.

EmpID	EmpName	DeptID	Salary	ProjectID
101	Rahul	1	60000	1
103	Amit	1	55000	1

2 rows in set (0.009 sec)

10. Design a relational schema for an Airline Reservation System and write 5 SQL queries demonstrating joins, sub-queries, and views.

```
mysql> DESC Flight;
```

Field	Type	Null	Key	Default	Extra
FlightID	int	NO	PRI	NULL	
FlightName	varchar(50)	YES		NULL	
Source	varchar(50)	YES		NULL	
Destination	varchar(50)	YES		NULL	
Fare	decimal(10,2)	YES		NULL	

5 rows in set (0.017 sec)

```
mysql> Select * From Flight;
```

FlightID	FlightName	Source	Destination	Fare
101	IndiGo	Chennai	Delhi	5500.00
102	Air India	Hyderabad	Mumbai	4800.00
103	SpiceJet	Bangalore	Kolkata	6200.00

3 rows in set (0.006 sec)